Ali Youssef

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Education | Employment | Publications | Projects | Skills | Certifications | Languages

Education

University of Bristol — Bristol, United Kingdom
Doctor of Philosophy in Computer Science — Sep. 2026 – Present
Supervisors: Dima Damen.

University College London — London, United Kingdom
Master of Science in Robotics and Computation — Sep. 2022 – Sep. 2023
Supervisor: Francisco Vasconcelos.
Classification: Distinction.

University of Glasgow — Glasgow, United Kingdom
Bachelor of Engineering in Mechanical Engineering — Sep. 2016 – Jun. 2020
Classification: First-Class Honours.

Employment

Computer Vision Research Intern — May 2025 – Dec. 2025
Spatial Intelligence — London, United Kingdom

- Mentored three postgraduate students in collaboration with the University of Bristol during joint academic-industry dissertations, providing both theoretical and technical guidance.
- Developed a self-supervised monocular depth estimation framework using a multi-modal Joint Embedding Predictive Architecture (JEPA) for metric depth prediction across diverse settings.
- Employed state-of-the-art object detection and tracking methods for temporally consistent multi-object tracking in dynamic environments.
- Analysed emerging research trends, assessing their applications and alignment with short- and medium-term objectives.

Independent Researcher — Nov. 2023 – Jul. 2024
University College London — London, United Kingdom
In collaboration with MSc supervisor post-graduation:

- Extended thesis scope by integrating additional baselines and model configurations to further validate the proposed research methodology.
- Refined evaluation protocols and conducted comprehensive analysis across multiple benchmarks, ensuring reliable and accurate model assessment.
- Enhanced models' performance and computational efficiency through extensive hyperparameter optimisations, algorithm vectorisation, and high-performance computing frameworks (JAX).
- Research culminated in a paper published at the ECCV 2024 Map-free Visual Relocalization Workshop.

Mechanical Engineer Intern — Jun. 2019 – Jul. 2019
Porsche Centre Egypt — Cairo, Egypt

- Performed comprehensive vehicle diagnostics and initial inspections under senior engineers' supervision.
- Executed routine maintenance and service tasks in coordination with workshop teams to resolve vehicle faults.
- Verified and documented that all serviced vehicles met strict quality and safety compliance standards.

Publications

Conference Proceedings

[1] **A. Youssef** and F. Vasconcelos. "NeRF-Supervised Feature Point Detection and Description". In: *European Conference on Computer Vision (**ECCV**) Workshops*, 2024. Paper / Code.

Preprints

[1] **A. Youssef**. "VMatcher: State-Space Semi-Dense Local Feature Matching". *ArXiv preprint arXiv:2507.23371*, 2025. Paper / Code.

Projects

Optimising Feature Point Detection and Description Using Novel View Synthesis Mar. 2023 – Sep. 2023
University College London London, United Kingdom

- Leveraged Neural Radiance Fields (NeRFs) to synthesise realistic multi-view training datasets containing both indoor and outdoor scenes with known camera parameters and depth maps.
- Adapted feature point detectors and descriptors to train on NeRF-synthesised data using perspective projection geometry for supervision, improving generalisability and reducing convergence time by 46%.
- Achieved competitive performance on geometric vision tasks (pose estimation, point cloud registration, and homography estimation) while requiring 97% less training data.

Data-Centric Wind Power Forecasting via Recurrent Neural Networks Oct. 2019 – Apr. 2020
University of Glasgow Glasgow, United Kingdom

- Applied Recurrent Neural Networks (RNNs) for wind turbine power forecasting.
- Evaluated outlier detection algorithms (Isolation Forest, Elliptic Envelope, and DBSCAN) to demonstrate the impact of data preprocessing on regression performance, with Isolation Forest proving most effective.
- Benchmarked Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, with GRU maintaining high accuracy while reducing computational overhead by 25%.

Skills

Programming Languages	Python | C++ | MATLAB
Deep Learning Frameworks	PyTorch | Triton | JAX
Robotics	Robot Operating System (ROS)
Cloud Computing	Amazon Web Services (AWS)

Certifications

AWS Solutions Architect – Associate (SAA-C02). Credential. Aug. 2022 – Aug. 2025

Languages

English — Arabic